

Q. $x = 10111$ Out put

bin2dec 31

bin2dec('1101111')

dec2bin(24) 11000

syms x

$y = \cos(x)$

diff(y)

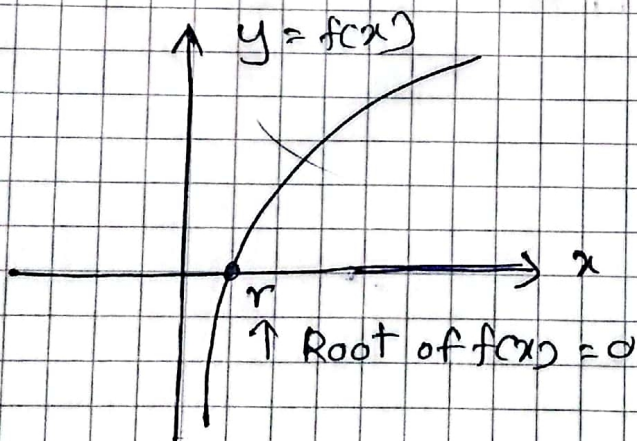
$-\sin(x)$

Chapter 02 Roots of Non linear Equation.

~~f(x)~~

Let $f(x)$ be a function

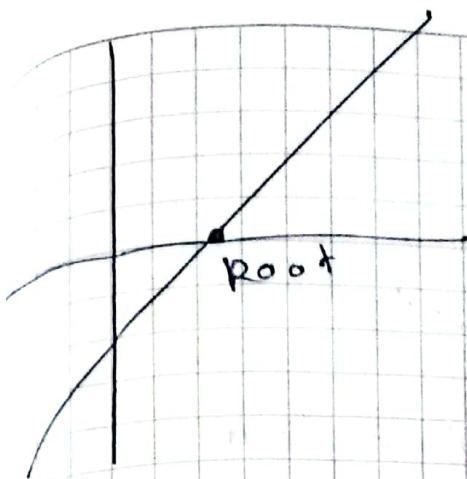
A number r satisfying $f(r) = 0$ is called a root of the equation $f(x) = 0$



Special cases

1. $f(x) = ax + b$, $a \neq 0$

$r = -\frac{b}{a}$ is the root of $f(x) = 0$



ii) $f(x) = a_2x^2 + a_1x + a_0, a_2 \neq 0$

There are two roots.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-a_1 \pm \sqrt{a_1^2 - 4a_2a_0}}{2a_2}$$

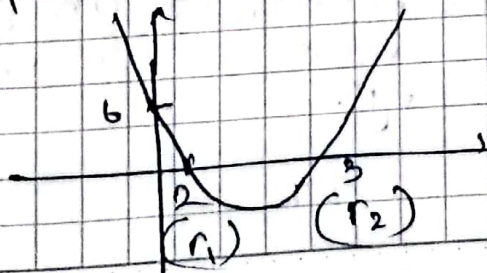
$$r_1 = \frac{-a_1 + \sqrt{a_1^2 - 4a_2a_0}}{2a_2}$$

$$r_2 = \frac{-a_1 - \sqrt{a_1^2 - 4a_2a_0}}{2a_2}$$

* Let

$$f(x) = x^2 - 5x + 6$$

$r_1 = 2$ $r_2 = 3$ are roots of $f(x) = 0$



Wise

$$P = 3x^2 + 2x - 5$$

$$MP = [3 \quad 2 \quad -5]$$

roots (P).

(iff) Let $f(x) = a_3x^3 + a_2x^2 + a_1x + a_0$

↑
A polynomial of degree 3

There are formulas,

Let $f(x) = a_nx^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0$; $a_n \neq 0$

when $n \geq 3$ there are no formulas to directly find the roots of the eq^y $f(x) = 0$

Also, there are no direct methods to find a root of a transcendental eq^y such as $ex = x$, $\tan x = x$, $xe^x - 1 = 0$, $\cos x - xe^x = 0$, $e \neq 0$

We need some approximation methods
Iterative

Iterative methods.

These methods are based on successive approximations for a desired root

Starting with one or more initial approximations a sequence of ~~app~~ approximations $\{x_n\}$ is obtained s.t. eg^y

$$\lim_{n \rightarrow \infty} x_n = r, \text{ when } r \text{ is an exact root of a desired eq^y}$$

iterative method for finding a root of the eqⁿ

$f(x) = 0$ takes the form

$$x_{n+1} = \phi(x_n) \text{ for all } n=0, 1, 2, 3, 4, \dots$$

x_0 is called the initial approximation.

$\phi(x)$ is called the iteration function.

$$x_{n+1} = \phi(x_n); n=0, 1, 2, 3, \dots \text{ requires}$$

exactly one initial value. Such an ~~method~~ iteration

method is called a one point

iteration method.

$$x_{n+1} = \phi(x_{n-1}, x_n) \quad x_n = \phi(x_0, x_1)$$

$$; n=0, 1, 2, 3, \dots$$

required two initial values.

Such a iteration method called two point iteration method.

$$x_n \rightarrow r$$

$$f(x_n) \rightarrow 0$$

$$\text{Error} \geq |f(x_k) - 0| = f(x_k)$$

$$f(x_n) = 0$$

$$f(x_{n+1} - x_n) < a$$

Stopping Criterion

Since we can't perform infinite number of iterations, we have to stop iteration when

$$\rightarrow |f(x_n)| \leq \epsilon \quad \text{or} \quad |x_{n+1} - x_n| \leq \epsilon$$

Exact value of $f(x) = 0$ when $x = r$, the

exact rule:

$$\text{error is } = |f(x_n) - f(r)|$$

$$= |f(x_n) - 0|$$

$$= |f(x_n)|$$

$$\text{Error: } \cancel{x_n} = |x_{n+1} - x|$$

$$\begin{aligned} |x_{n+1} - x| &= |x_{n+1} - x_n + x_n - x| \\ &\leq |x_{n+1} - x_n| + |x_n - x| \\ &\leq |x_{n+1} - x| + |x_n - x| \\ &< \epsilon \end{aligned}$$

ϵ has to be given

ϵ - Tolerance

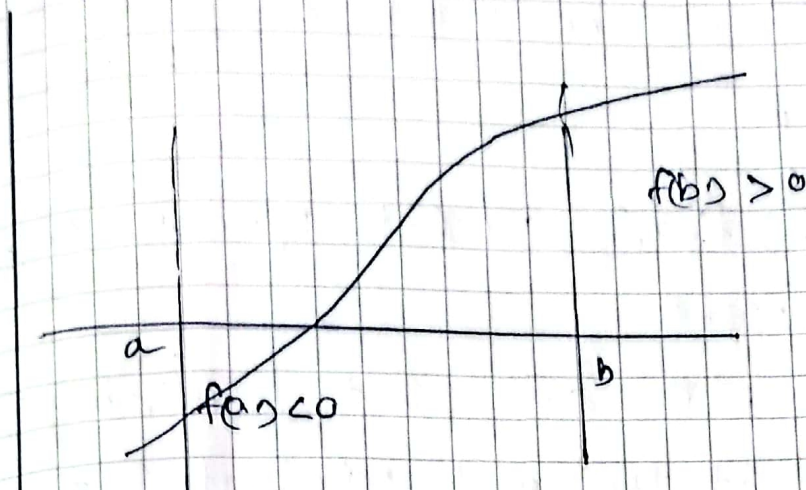
$= \epsilon_0$

Bisection Method:

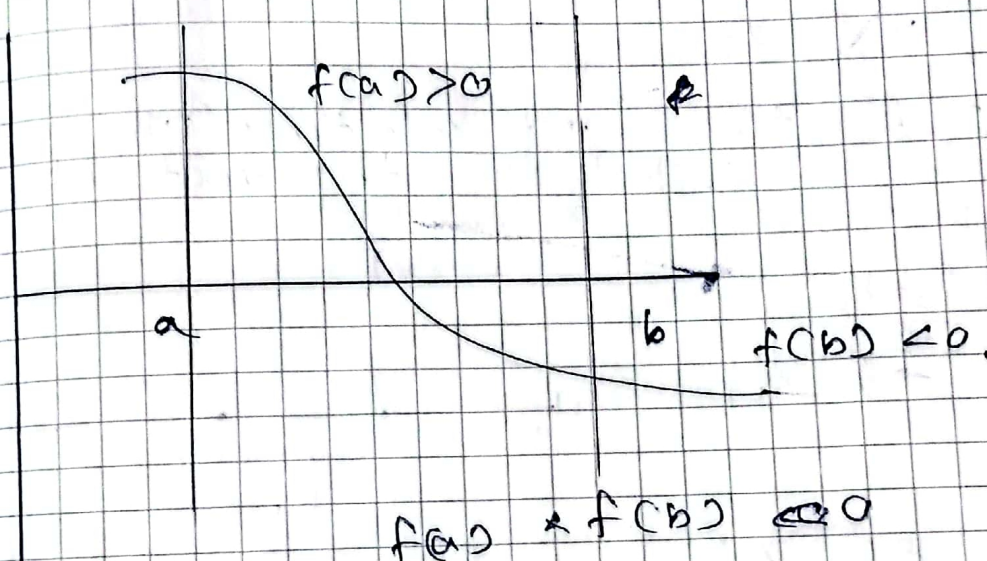
The Bisection method based on the following theorem. we called Intermediate Value theorem (IVT)

Let, f be a continuous function over the closed interval $[a, b]$

Suppose that $f(a)$ and $f(b)$ have ~~per~~ opposite signs
 then there exist $r \in [a, b]$ s.t. $f(r) = 0$



$$f(a) \times f(b) < 0.$$



$$f(a) \times f(b) < 0$$

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γ

Suppose that a root γ of $f(x) = 0$ lies in the interval $I_0 = (a_0, b_0) = (a_0, b_0)$

$$I_0 = (a_0, b_0) = (a_0, b_0)$$

1. Bisection I_0 at the point $x_1 = \frac{a_0 + b_0}{2}$

$$\therefore \gamma \approx x_1$$

2. If $f(a_0) f(x_1) < 0$ let $I_1 = (a_0, x_1) = (a_1, b_1)$

If this happens

$$f(x_1) \cdot f(b_0) < 0, \text{ let } I_{1,2} = (x_1, b_0) = (a_1, b_1)$$

3. Bisection I_1 at the mid point of I_1

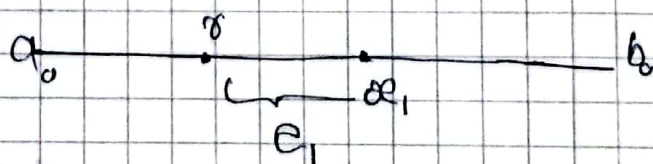
$\therefore x_2 \approx$ the mid point of I_1

$$\gamma \approx x_2$$

* Keep bisectioning and selecting (choosing) an interval including the root until a given accuracy is met

$$x_1 = \frac{a_0 + b_0}{2}$$

$$\Rightarrow e_1 = |x - x_1|$$



$$e_1 = |x - x_1| \leq \frac{1}{2} (b_0 - a_0)$$

$$x_2 = \frac{a_1 + b_1}{2} \Rightarrow e_2 = |x - x_2| \leq \frac{1}{2} (b_1 - a_1)$$

$$x_3 = \frac{a_2 + b_2}{2} \Rightarrow e_3 = |x - x_3| \leq \frac{1}{2} (b_2 - a_2)$$

or

Wise

At the n^{th} iteration

$$e_n = |x - x_n| \leq \frac{1}{2} |b_{n-1} - a_{n-1}| \quad \text{--- (1)}$$

Now (1) implies

Suppose for a given error ϵ

$$\frac{b_0 - a_0}{2\epsilon} \leq \epsilon$$

$$\Rightarrow (b_0 - a_0) \leq \varepsilon \cdot 2^n$$

$$\log(b_0 - a_0) \leq \log(e \cdot 2^n)$$

$$\Rightarrow \log e + n \log 2$$

$$\Rightarrow n \geq \frac{\log(b_0 - a_0) - \log \varepsilon}{\log 2}$$

⇒ next Wise page.

$$1.5 \rightarrow 2$$

$$1.1 \rightarrow 2$$

$\lceil x \rceil$ ~~$\lfloor x \rfloor$~~ = ceiling function is a integer such that

$$\lceil x \rceil \geq x$$

smallest integer $\geq x$

$$\lceil 1.5 \rceil = 2$$

$$\lceil 1.1 \rceil = 2$$

$$\lceil 3 \rceil = 3$$

~~x~~

~~$\lceil x \rceil$~~ $\lfloor x \rfloor$ = floor function is a integer.

$$\lfloor x \rfloor \leq x$$

$\lfloor x \rfloor$ = largest integer $\leq x$

$$\lfloor 1.3 \rfloor = 1$$

$$\lfloor -2.3 \rfloor = -3$$

$$\lfloor 5 \rfloor = 5$$

$$\text{set } n = \left\lceil \frac{\log(b_0 - a_0) - \log \epsilon}{\log 2} \right\rceil$$

Let $\epsilon = 10^{-2}$ ~~$\epsilon = 0.01$~~ $\epsilon = 0.01$ and let $a_0 = b_0 = 1$

$$\frac{\log(b_0 - a_0) - \log \epsilon}{\log 2} = \frac{\log(1-1) - \log 10^{-2}}{\log 2}$$

$$= \log 1 + 12 \log 10 = 6.6438561.$$

$$\therefore n = \left\lceil \frac{\log 2}{6.6438561} \right\rceil = 7$$

ϵ	10^{-2}	10^{-3}	10^{-4}	10^{-5}	10^{-6}	10^{-7}
n	7	10	14	17	20	24

Example

Consider the eqⁿ

$$x^3 - 2x - 1 = 0$$

Show that this eqⁿ has a root between 1 and 2. Find it correct to 4 decimal places.

$$\text{Let } f(x) = x^3 - 2x - 1$$



$$f(1) = 1^3 - 2 \times 1 - 1$$

$$= 1 - 2 - 1 = -2 < 0$$

$$f(2) = 2^3 - 2 \times 2 - 1$$

$$= 8 - 4 - 1 = 3 > 0$$

$$\therefore f(1) \cdot f(2) < 0$$

$\therefore$ There must be a root between 1 and 2.

$$x = 0 : 0.01 : 20;$$

$$y = x.^3 - 2 * x - 1;$$

plot(x, y)

$$x_1 = \frac{1+2}{2} = \frac{3}{2} = 1.5$$

$$f(x_1) = (1.5)^3 - 2 \times 1.5 - 1 < 0$$

Since $f(x_1) = f(1.5) < 0$, the root must be between

$$x_2 = \frac{x_1 + 2}{2} = \frac{1.5 + 2}{2} = 1.75$$

$$f(x_2) = (1.75)^3 - 2 \times 1.75 - 1 > 0$$

Since $f(x_1) \cdot f(x_2) < 0$, the root must be between x_1 and x_2 .

We need to calculate correct to 4 decimal places.

Note: Let x be a number and

x^* be an approximation for x .

If $|x - x^*| \leq 0.5 \times 10^{-n}$, then

x^* represents x accurate to n decimal places (n significant digits).

Let $\epsilon = 0.5 \times 10^{-4}$
 $= 0.00005$

$$|x_{k+1} - x_k| \leq \epsilon$$

$$|x_{n+1} - x_n| \leq \epsilon$$

$$|x_1 - x_2| = |x_2 - x_1| = |1.75 - 1.5| = 0.25$$

$$0.25 \geq 0.00005$$

$$|x_3 - x_2| = |1.625 - 1.5| = 0.125 > 0.00005$$

After 15 iteration answer for
 $x \approx 1.6180$ $0.00003 < 0.00005$

from the matlab program we have,

$$|x_{15} - x_{14}| = 0.00003 < 0.00005$$

Q4-w> Show that $x^2 - x = 0$ has a root
 between $\frac{1}{2}$ and $\frac{3}{2}$ and determine it

Show that $x e^x - 1 = 0$ has a root
 between 0.5 and 1.0. determine it
 correct to 3 decimal places.


```

f = @(x) x.^3 - 2*x - 1;
a = 1; b = 2;
x0 = a;
x1 = (a+b)/2;

n = 1;
E = 0.00005;
Err = abs(x0 - a);
L = [n a b x1 Err];
fprintf('n\t a\t b\t x1\t Err\n');
format spec = '%d %1.4f %1.4f %1.4f %1.4f\n';
fprintf(format spec, L);

while (abs(x1 - x0) >= E)
    if (f(a) * f(x1) < 0)
        b = x1;
        x1_Old = b;
    else
        a = x1;
        x1_Old = a;
    end;
    n = n + 1;
    x1 = (a+b)/2;
    x0 = x1_Old;
    Err = abs(x1 - x0);
    L = [n a b x1 Err];
    fprintf
    fprintf(format spec, L);
end;

```

$x^2 - x = 0$ between $\frac{1}{2}$ and $\frac{3}{2}$

$$f(x) = x^2 - x$$

$$f\left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)^2 - \frac{1}{2}$$

$$= \frac{1}{4} - \frac{1}{2} = -\frac{1}{4} < 0$$

$$f\left(\frac{3}{2}\right) = \left(\frac{3}{2}\right)^2 - \frac{1}{2}$$

$$= \frac{9}{4} - \frac{1}{2} = \frac{18 \times 9 - 2}{4} = \frac{142}{4} = 71 \frac{1}{2} > 0$$

$$f\left(\frac{1}{2}\right) \cdot f\left(\frac{3}{2}\right) < 0$$

There exist root between $\frac{1}{2}$ and $\frac{3}{2}$

$$f(x) = xe^x - 1 = 0$$

0.5 and 1.0.

$$f(0.5) = 0.5 \times e^{0.5} - 1$$

$$= -0.175644 < 0$$

$$f(1.0) = 1 \times e^1 - 1$$

$$= 1.71828 > 0$$

$$\begin{array}{c} f(x) \\ \hline x_0 \quad x_2 \quad x_1 \end{array}$$

$$f(0.5) \cdot f(1.0) < 0$$

∴ There exist root between 0.5 and 1.

So $x_0 = 0.5$ and $x_1 = 1.0$.

$$x_2 = \frac{x_1 + x_0}{2} = \frac{1.0 + 0.5}{2} = \frac{1.5}{2} = 0.75$$

$$f(x_2) = 0.75 \times e^{0.75} - 1$$

$$= 1.11700 > 0$$

~~So~~ the $f(x_2) \cdot f(x_0) < 0$ So the root must be between x_0 and x_2 .

MATLAB Built in functions for finding roots

the `fzero` command can be used to solve non linear eq^y (finding roots).

It needs to know approximate where the solution (root) is

$$x = \text{fzero}(f(x), x_0)$$

↑ solution ↑ non linear function ↑ Approximation solution

If $x = \cos x$
 $x - \cos x$
 $f(x) = x - \cos x$

$$f(x) = 8 - 4.5(x - \sin x)$$

$$x_0 = ? \quad f(x) = 0$$

↑
we want to solve this.

code.

~~f = 0(x)~~

$$f = @(x) 8 - 4.5 * (x - \sin(x))$$

Draw the graph

$$x = -2: 0.01: 10;$$

$$y = f(x);$$

$$\text{plot}(x, y)$$

$$\text{fzero}(f, 2)$$

ans =

2.4305.

format long

$$\text{fzero}(f, 2)$$

ans =

~~2.4305~~ 2.43046574172360

Wise

vpa → variable position

digits(100);

vpa(fzero(f, 2))

2.4300045 ~ ~ ~

digits(40);

vpa(fzero(f, 2))

Roots command

This can be used to find root of a polynomial eqⁿ

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 \leftarrow a_n$$

nth polynomial

In Matlab,

$$p = [a_n \ a_{n-1} \ a_{n-2} \ \dots \ a_1 \ a_0]$$

~~P = f~~

ex.: $p = x^3 - 2x - 1$

$$\overset{+3}{x} + \overset{+2}{0x^2} + \overset{+1}{-2x} - 1$$

$$p = [1 \ 0 \ -2 \ -1] : x^3 + 0 \cdot x^2 - 2x - 1$$

roots(p).

$$\text{Ans} = 1.6180339887495$$

$$-1.0000000000000$$

$$0.6180339887495$$

zero method.

$$f = p(x) = x^3 - 2x - 1$$

$$x = \text{fzero}(f, 1)$$

$$x = 1.6180339887495$$

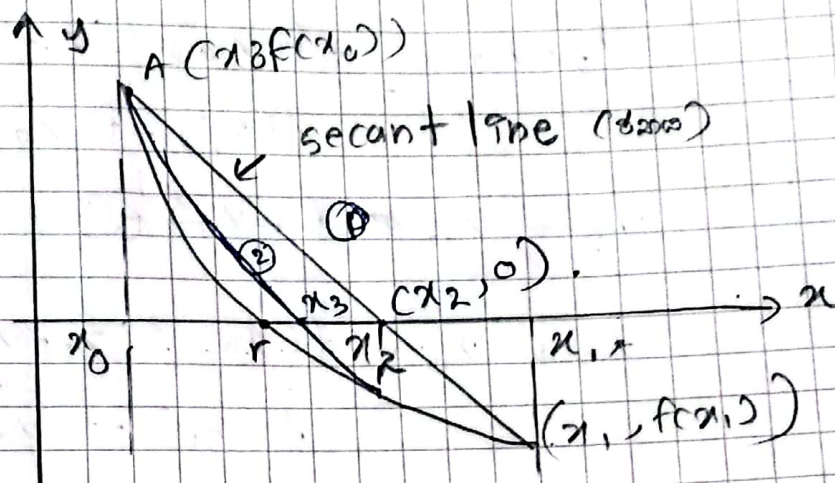
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Regula - Falsi Method / Method of false position

Let x_0 and x_1 be two initial approximations to the root r of the equation $f(x)=0$

We also assume that the root that is r lies between x_0 and x_1 ,

$$\text{so } f(x_0) \cdot f(x_1) < 0$$



x_2 : The next approximation to the root r .

This is the point at which the secant line

through the points a and b cuts the x axis

The equation of the secant line AB is

$$\frac{y - f(x_0)}{x - x_0} = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

$$y = \left(\frac{f(x_1) - f(x_0)}{x_1 - x_0} \right) (x - x_0) + f(x_0)$$

$$y = \left(\frac{f(x_1) - f(x_0)}{x_1 - x_0} \right) (x - x_0) + f(x_0)$$

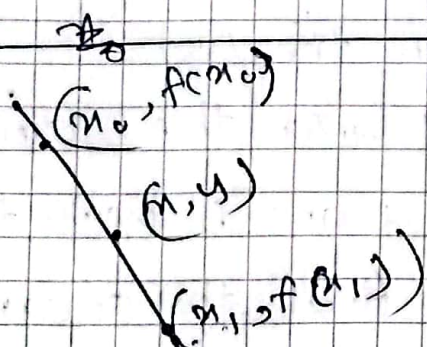
Next we want to find the x_2 . To get this we substitute $x = x_2$ and $y = 0$ in (1)

$$\therefore 0 = \frac{f(x_1) - f(x_0)}{(x_1 - x_0)} (x_2 - x_0) + f(x_0)$$

$$x_2 - x_0 = \frac{-f(x_0)(x_1 - x_0)}{f(x_1) - f(x_0)}$$

$$x_2 = x_0 + \frac{-f(x_0)(x_1 - x_0)}{f(x_1) - f(x_0)}$$

$$x_2 = x_0 - \frac{f(x_0)(x_1 - x_0)}{f(x_1) - f(x_0)}$$



$$\frac{y - f(x_1)}{x - x_1} = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

$$y = \frac{f(x_1) - f(x_0)}{x_1 - x_0} (x - x_1) + f(x_1)$$

$$\therefore 0 = \frac{f(x_1) - f(x_0)}{(x_1 - x_0)} (x_2 - x_1) + f(x_1)$$

$$x_2 = \frac{-f(x_1)(x_1 - x_0)}{f(x_1) - f(x_0)} + x_1$$

↑ we need this form

Wise

$$\frac{1}{x_0} \quad \frac{1}{x_3} \quad \frac{1}{x_2} \quad \frac{1}{x_1}$$

$$x_0 \approx x_1 - T_0$$

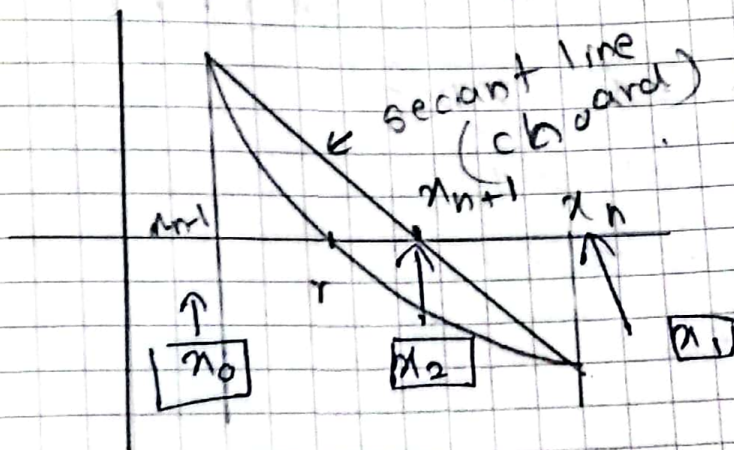
$$x_0 \approx x_2 - T_1$$

$$x_0 \approx x_3 - T_2$$

$$x_3 \approx x_2 - \frac{(x_2 - x_1) f(x_2)}{f(x_2) - f(x_1)}$$

$$x_4 \approx x_3 - \frac{(x_3 - x_2) f(x_3)}{f(x_3) - f(x_2)}$$

Suppose that x_{n-1} and x_n are two approximations to the root r of the $f(x) = 0$.



The general we have

$$x_{n+1} \approx x_n - \frac{(x_n - x_{n-1}) f(x_n)}{f(x_n) - f(x_{n-1})}$$

For all $n = 1, 2, 3, 4, 5$

In this method, at each statement step, we have to find the interval that includes the root

Example

Consider $f(x) = x^3 - 5x + 1 = 0$

(i) Show that $f(x) = 0$ has a root between 0 and 1

(ii) Determine it using R.F.M

~~$f(x_0) = 0^3 - 5 \times 0 + 1$
 $= 0 - 0 + 1$
 $= 1 > 0$
 $f(x_1) = 1^3 - 5 \times 1 + 1$
 $= 1 + 1 - 5$
 $= -3 < 0$~~

$f(x_0) = 0^3 - 5 \times 0 + 1$
 $= 1 > 0$

$f(x_1) = 1^3 - 5 \times 1 + 1$
 $= 1 + 1 - 5$
 $= -3 < 0$

$\therefore$ In fact y is a continuous function

$\therefore$ There exists a root of $f(x) = 0$ between x_0 and x_1 , $x_0 = 0, x_1 = 1$

Wise

(if) To determine it, we use (RPM).

$$x_2 = x_1 - \frac{f(x_1)(x_1 - x_0)}{f(x_1) - f(x_0)}$$

$$= 1 - \frac{(-3) \times (1 - 0)}{(-3 - 9)} \quad \text{---}$$

$$= 1 - \frac{(-3) \times 1}{-4} \quad \text{---}$$

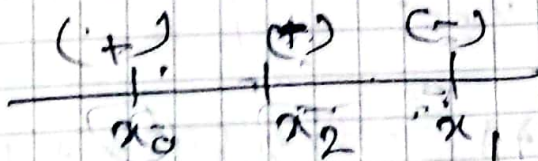
$$= 1 + \frac{3}{4} \quad \text{---}$$

$$= 1 + \frac{3}{4}$$

$$= 1 + 0.75$$

$$= 1.75$$

$$f(x_2) = -0.2344 < 0$$



The root exists between $x_0 = 0$ and $x_2 = 0.25$.

$$x_3 = x_2 - \frac{f(x_2)(x_2 - x_1)}{f(x_2) - f(x_1)}$$

$$= 0.25 - \frac{(-0.2344)(0.25 - 1)}{-0.2344 + 3} = 0.20253$$

Wise

$$\begin{array}{c} \nearrow x_{n-1} \quad x_n \\ (+) \quad (+) \quad (-) \\ \hline x_0 \quad x_1 \quad x_2 \end{array}$$

$$x_3 = x_2 - \frac{(x_2 - x_0) f(x_2)}{f(x_2) - f(x_0)}$$

$$= 0.25 - \frac{(0.25 - 0)(-0.2344)}{(-0.2344 - 1)}$$

$$= 0.25 - \frac{0.25 \times -0.2344}{1.2344}$$

$$= 0.25 + \frac{0.0586}{1.2344}$$

$$= 0.25 + 0.0475$$

$$= 0.2974$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$f(x_3) = (0.2974)^3 - 5 \times 0.2974 + 1$$

$$= 0.0263 - 1.487 + 1$$

$$= -0.4607$$

$$f(x_3) < 0$$

$$x_4 = \frac{x_3 - (x_3 - x_2) \frac{f(x_3) - f(x_2)}{f(x_3) - f(x_2)}}{f(x_3) - f(x_2)}$$

$$= \frac{0.2974 - (0.2974 - 0.25) \times \frac{-0.2344}{-0.4607 - 0.2344}}{-0.4607 - 0.2344}$$

$$= 0.2974 + \frac{0.0474 \times -0.2344}{0.2818}$$

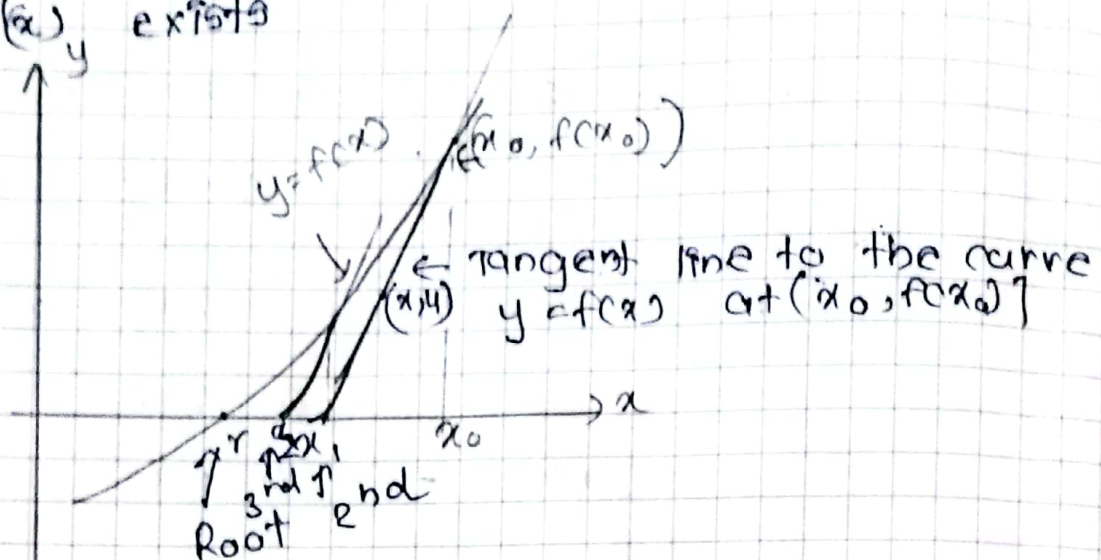
$$= 0.2974 - 0.0394$$

$$= -0.25847$$

Newton - Rapson Method / Newton method.

* Let f be the function which is continuous and differentiable

$f(x)$ exists



Let r be the root of $f(x) = 0$

Let x_0 be an initial approximation for
 $r \approx x_0$

first approximation : $x_0 \approx r$

second approximation : $x_1 \approx r$

! The point at which the tangent line to $y = f(x)$ at $(x_0, f(x_0))$ cuts the x axis.

Third approximation: for r : x_2 : The point at which the tangent line to $y = f(x)$ at $(x_1, f(x_1))$ cuts the x axis.

Wise

n^{th} approximation for x : x_{n-1} : The point at which the tangent line to $y=f(x)$ at $(x_{n-2}, f(x_{n-2}))$ cuts the x axis:

$f'(x_0)$: The slope of the tangent line to $y=f(x)$ at $(x_0, f(x_0))$

The eqⁿ of the tangent line to $y=f(x)$ at $(x_0, f(x_0))$
 $y = (x - x_0) f'(x_0) + f(x_0)$

$$y = mx + c$$

$$y = f'(x_0)x + c$$

$$f(x_0)$$

$$y = f'(x_0)x + f(x_0) - f(x_0)x_0$$

$$= (x - x_0) f'(x_0)$$

$$= [f'(x_0) - f(x_0)]x + f(x_0)$$

$$f(x_0) = f'(x_0)x_0 + c$$

$$c = f(x_0) - f'(x_0)x_0$$

$$y = (x - x_0) f'(x_0) + f(x_0)$$

$$\text{At } (x, 0)$$

$$0 = (x - x_0) f'(x_0) + f(x_0)$$

$$(x - x_0) f'(x_0) = -f(x_0)$$

$$x - x_0 = -\frac{f(x_0)}{f'(x_0)}$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Wise

Similarly,

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} \quad \text{if } f'(x_1) \neq 0$$

At n^{th} Approximation, we have.

$$x_{n+1} = x_{n-2} - \frac{f(x_{n-2})}{f'(x_{n-2})}$$

OR

$$x_n = x_{n-1} - \frac{f(x_{n-1})}{f'(x_{n-1})} \Rightarrow (n+1)^{\text{th}} \text{ approximation}$$

for $n=1, 2, 3$.

Stopping Criterion.

(i) Relative error

$$\left| \frac{x_n - x_{n-1}}{x_{n-1}} \right| \leq \epsilon = \text{TOL}$$

$$(ii) \left| f(x)_{n-1} \right| \leq \epsilon = \text{TOL}$$

Example 01

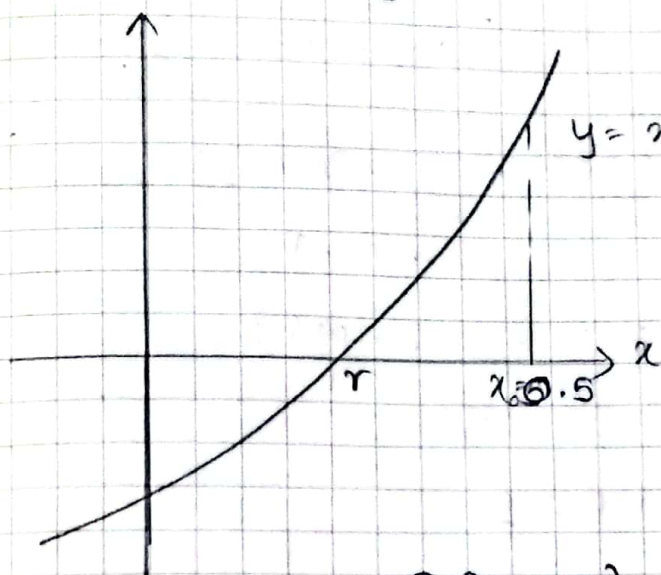
consider the eqⁿ, $xe^x - 1 = 0$.

$$x = 0:0.001:2;$$

$$y = x * \exp(x) - 1;$$

$$\text{plot}(x, y);$$

$$fzero(f, 0.5)$$



$$y = x e^x - 1$$

$$y = x e^x - 1 = f(x)$$

$$f'(x) = x e^x + e^x$$

$$= (1+x) e^x$$

$$fzero(f, 0.5)$$

$$\text{Let } x_0 = 0.5$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_0 = 0.5$$

$$f'(x_0) =$$

$$= 0.5 - \frac{e^{-0.47308} (-0.1756)}{2.47308} = (1+0.5) e^{0.5}$$

$$= 1.5 \times 1.64872$$

$$= 2.47308$$

$$= 0.5 + \frac{0.1756}{2.4731}$$

$$f(x) = x e^x - 1$$

$$= 0.5 \times e^{0.5} - 1$$

$$= -0.1756$$

$$= \underline{0.57102}$$

Wise

$$x_3 = x_2 - f(x_2) / g(x_2)$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$f(x_1) = (0.57102)e^{1.4700} - 1$$

$$= (1.4700)(0.57102)$$

$$= 0.57102 - \frac{f(0.57102)}{f'(0.57102)} = 0.010446$$

$$= 0.57102 - \frac{0.010446 f(x_1)}{2.780754} = \frac{(1 + 0.57102)}{e^{1.4700}}$$

$$= 0.57102 - 3.864487$$

$$= 0.567155$$

$$= \underline{\underline{0.5679}}$$

$$= \frac{1.57102}{1.4700}$$

$$= 2.7807054$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$= \frac{0.5679}{e^{1.4700}} = 1.48332$$

$$= 0.5679 - \frac{1.5670851}{2.763475}$$

$$= 0.5679 - 0.5670704$$

$$= -\cancel{0.0001} 0.0003$$

Secant method

~~NR~~

In NR (Nurderupton method) we have to calculate $f'(x)$.

This is a major difficulty of the NR. To overcome it we use

$$f'(x) = \frac{f(x_{n-1}) - f(x_n)}{x_{n-1} - x_n}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x) - f(x-h)}{h}$$

The NR formula becomes ~~$x_{n+1} = x_n - f(x_n)$~~

$$x_{n+1} = x_n - \frac{f(x_n)(x_{n-1} - x_n)}{f(x_{n-1}) - f(x_n)}$$

for $n = 1, 2, 3, 4, 5, \dots$

We need two initial approximations for the root

Here we don't consider the interval where the root exist at each steps

Use

Consider the ~~the~~ previous example

$$xe^x - 1 = 0$$

$$\text{Set } x_0 = 5 \text{ and } x_1 = 1$$

$$x_{n+1} = x_n - \frac{f(x_n)(x_{n-1} - x_n)}{f(x_{n-1}) - f(x_n)}$$

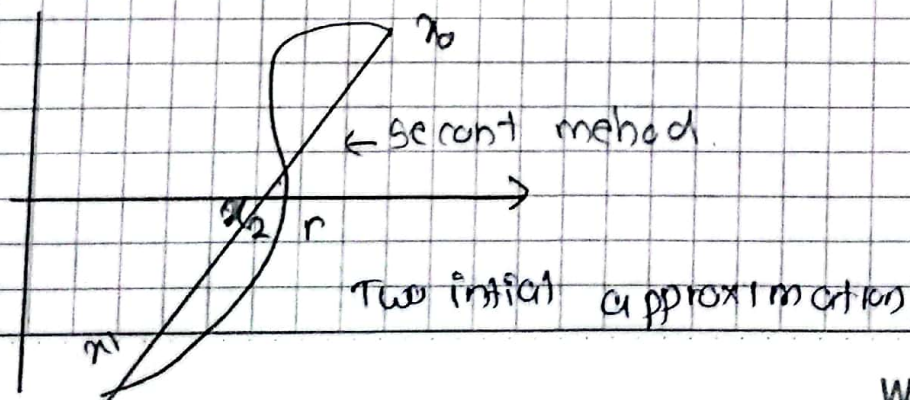
$$x_2 = x_1 - \frac{f(x_1)(x_0 - x_1)}{f(x_0) - f(x_1)}$$

$$x_3 = x_2 - \frac{f(x_2)(x_1 - x_2)}{f(x_1) - f(x_2)}$$

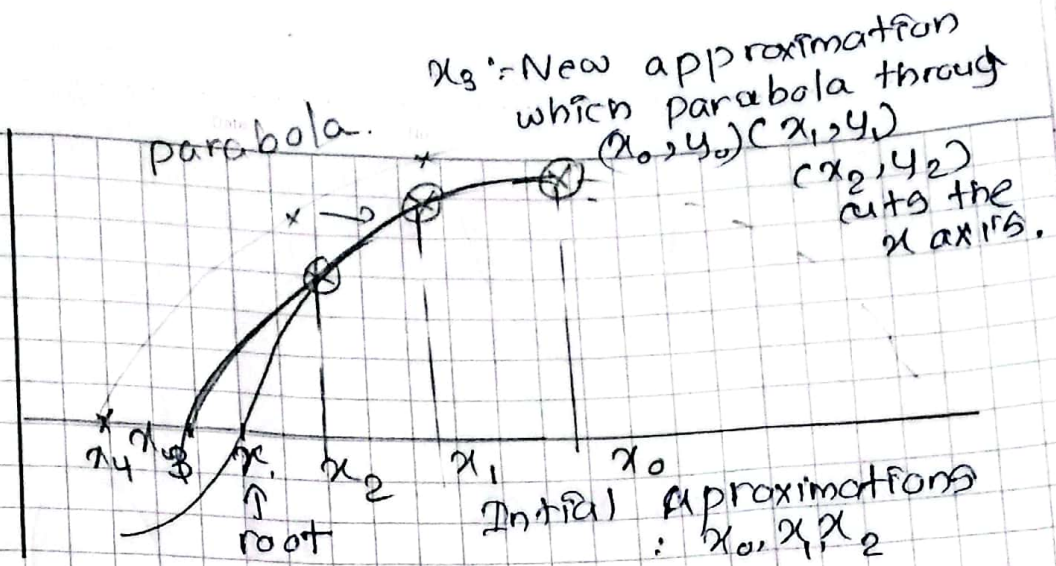
$$x_4 = x_3 - \frac{f(x_3)(x_2 - x_3)}{f(x_2) - f(x_3)}$$

Müller method

The Müller method is an extension of the secant method. We have



Wise



Here we need three initial approximation.

The Müller method can be used to approximate a root of real function.

Here we start with a real initial approximations to find the complex root.

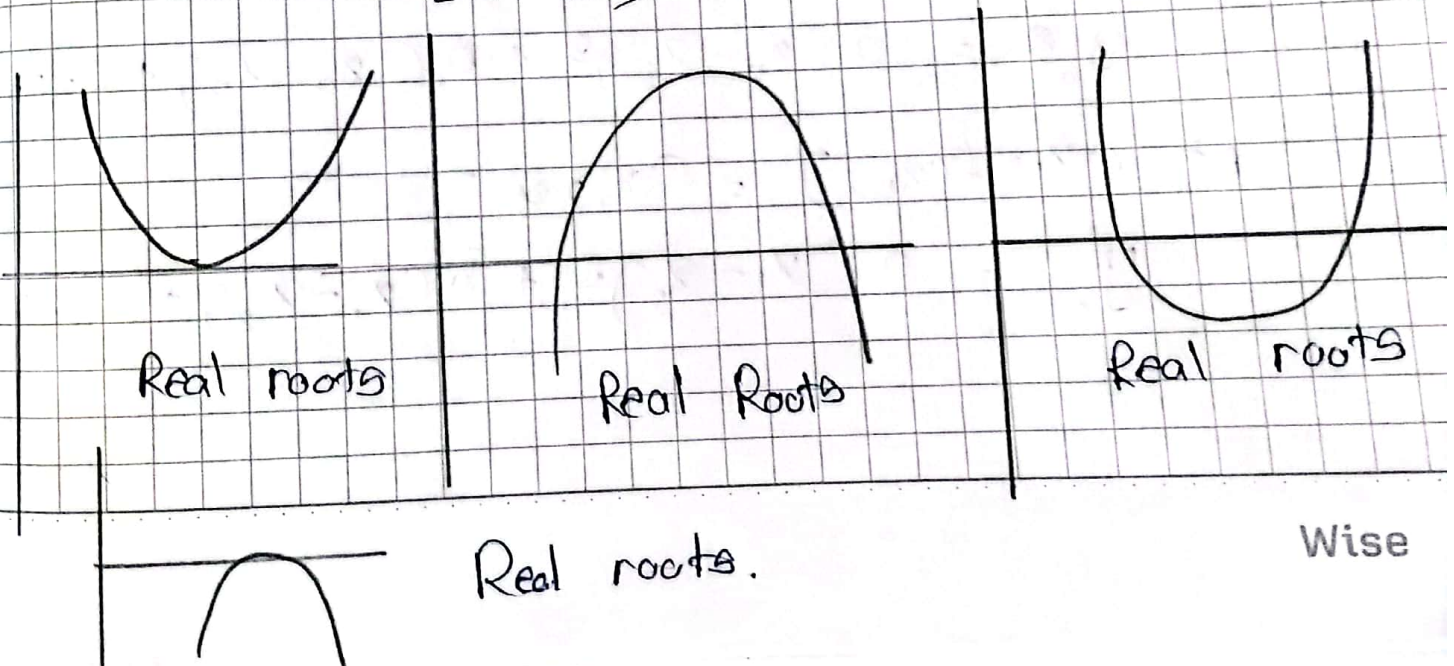
x_0, x_1, x_2

$[x_1] [x_2] [x_3]$

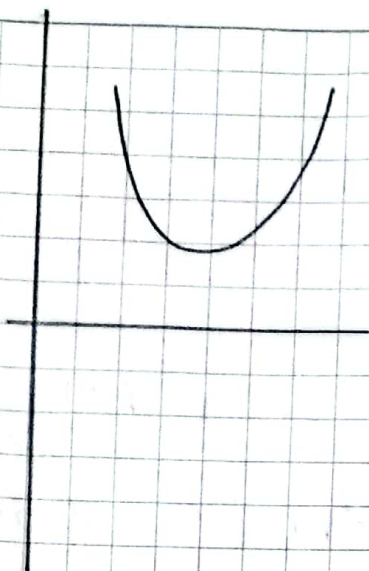
$[x_2] [x_3] [x_4]$

11/10/2023

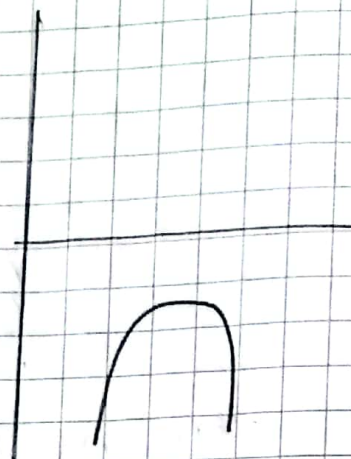
Step 1 : find the parabola passing through the points $(x_0, f(x_0))$ $(x_1, f(x_1))$ and $(x_2, f(x_2))$



Wise



complex roots



complex roots

Suppose that the equation of the parabola is given by

$$y(x) = a(x - x_2)^2 + b(x - x_2) + c$$

$$= Ax^2 + Bx + C$$

We must determine a , b and c using the points $(x_0, f(x_0))$, $(x_1, f(x_1))$ and $(x_2, f(x_2))$

$$A + (x_0, f(x_0)) = (x_0, y_0)$$

$$y_0 = a(x_0 - x_2)^2 + b(x_0 - x_2) + c = f(x_0) - c$$

$$A + (x_1, f(x_1)) = (x_1, y_1)$$

$$y_1 = a(x_1 - x_2)^2 + b(x_1 - x_2) + c = f(x_1) - c$$

$$At (x_2, f(x_2)) = (x_2, y_2)$$

$$y_2 = c = f(x_2) \quad \text{--- (3)}$$

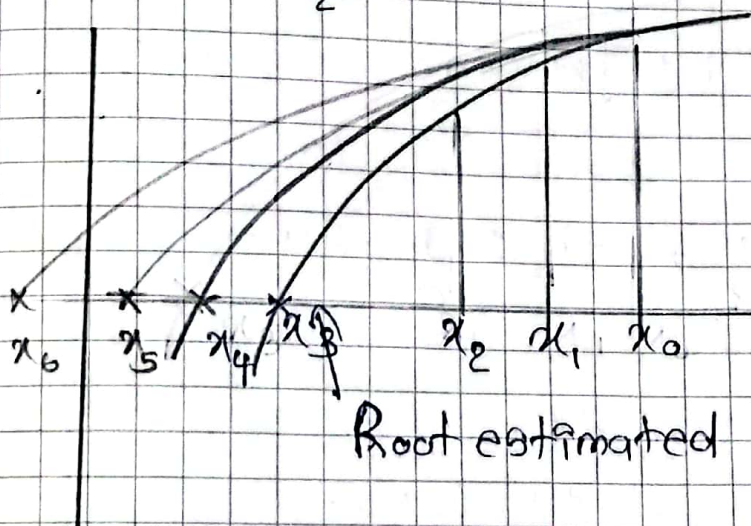
From (1) and (2), we get

$$a(x_0 - x_2)^2 + b(x_0 - x_2) = f(x_0) - c$$

$$a(x_1 - x_2)^2 + b(x_1 - x_2) = f(x_1) - c$$

$$\begin{bmatrix} (x_0 - x_2)^2 & (x_0 - x_2) \\ (x_1 - x_2)^2 & (x_1 - x_2) \end{bmatrix} \text{ or } \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} f(x_0) - c \\ f(x_1) - c \end{bmatrix}$$

where $c = f(x_2)$



$$(1) x_0, x_1, x_2 \Rightarrow x_3$$

$$(2) x_1, x_2, x_3 \Rightarrow x_4$$

$$(3) x_2, x_3, x_4 \Rightarrow x_5$$

$$(4) x_3, x_4, x_5 \Rightarrow x_6$$

$$(n) x_{n-1}, x_n, x_{n+1} \Rightarrow x_{n+2}$$

NOTE

Consider the quadratic equation

$$ax^2 + bx + c = 0 \quad (*)$$

Set $x = \frac{1}{t}$ where $t \neq 0$

Then we have from (*)

$$a\left(\frac{1}{t}\right)^2 + b\left(\frac{1}{t}\right) + c = 0$$

$$c + b t + a t^2 = 0$$

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\frac{1}{x} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{2a}{-b \pm \sqrt{b^2 - 4ac}}$$

$$\boxed{x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}}$$

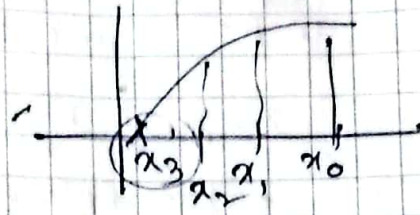
Consider the Quadratic equation

$$y(x_3) = 0$$

Solving,
This gives x_3

$$y(x_3) = 0 \Rightarrow$$

$$a(x_3 - x_2)^2 + b(x_3 - x_2) + c = 0$$



$$x_3 - x_2 = \frac{-2c}{b \pm \sqrt{b^2 - 4ac}}$$

$$x_3 = x_2 + \frac{-2c}{b \pm \sqrt{b^2 - 4ac}}$$

There are two choices for x_3

$$\text{If } |b + \sqrt{b^2 - 4ac}| > |b - \sqrt{b^2 - 4ac}|$$

$$\text{then } x_3 = x_2 + \frac{-2c}{b + \sqrt{b^2 - 4ac}}$$

$$\text{If } |b - \sqrt{b^2 - 4ac}| > |b + \sqrt{b^2 - 4ac}|$$

$$\text{then } x_3 = x_2 + \frac{-2c}{b - \sqrt{b^2 - 4ac}}$$

Example 1.

Approximate a real root of ~~$x^3 + 7x^2 + 6x + 5 = 0$~~

$$x^3 + 7x^2 + 6x + 5 = 0$$

$$f(x_0) = 0 \neq 5$$

$$f(x_1) = 1 - 7 + 6 + 5 = 5$$

Use $x_0 = 0$, $x_1 = 1$, $x_2 = 2$ as initial approximations.

$$\text{Let } f(x) = x^3 - 7x^2 + 6x + 5.$$

Iteration 1: computing x_3 from x_1, x_2, x_0

$$y = (x - x_2)^2 a + b(x - x_2) + c$$

we must determine a, b and c

$$c = f(x_2) = f(2) = 2^3 - 7 \cdot 2^2 + 6 \cdot 2 + 5$$

$$= 8 - 28 + 12 + 5$$

$$= 25 - 28$$

$$= -3 //$$

$$\begin{bmatrix} (x_0 - x_2)^2 & (x_0 - x_2) \\ (x_1 - x_2)^2 & (x_1 - x_2) \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} f(x_0) - c \\ f(x_1) - c \end{bmatrix}$$

$$\begin{bmatrix} 4 & -2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 8 \\ 8 \end{bmatrix}$$

$2 \times 2 \quad 2 \times 1 \quad 1 \times 1$

$$Ax =$$

$$4a - 2b = 8$$

$$a - b = 8$$

$$4a - 4b = 32$$

$$2b = -24$$

$$b = -12$$

$$a = 8 - 12$$

$$= -4 //$$

$$\begin{bmatrix} 4a - 2b \\ a - b \end{bmatrix} = \begin{bmatrix} 8 \\ 8 \end{bmatrix}$$

$$\begin{bmatrix} 4a + 2b \\ a + b \end{bmatrix} = \begin{bmatrix} 8 \\ 8 \end{bmatrix}$$

$$2a + b = 4$$

$$a + b = 8$$

$$a = -4 //$$

$$b = 12 //$$

Wise

$$a + b = 8$$

$$2(2a + b) = 8$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax+by \\ cx+dy \end{pmatrix} \quad A = \begin{bmatrix} 4 & -2 \\ 1 & -1 \end{bmatrix}$$

$$\boxed{\begin{matrix} x \in \text{inv}(A) \cap B \\ x \in A/B \end{matrix}}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-12 \pm \sqrt{144 - 4 \cdot 1 \cdot (-3)}}{2 \cdot 1}$$

$$= \frac{-12 \pm \sqrt{144 + 12}}{2}$$

$$= \frac{-12 \pm \sqrt{156}}{2}$$

$$\frac{4 \pm 12}{2}$$

$$= \frac{-12 \pm 12.49}{2}$$

$$\frac{-12.49}{2} = -6.245$$

$$\textcircled{1} \quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-12 \pm \sqrt{144 - 4 \cdot 1 \cdot (-3)}}{2 \cdot 1}$$

$$= \frac{-12 \pm \sqrt{156}}{2}$$

$$= \frac{-12 \pm 12.49}{2}$$

$$\textcircled{2} \quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-12 \pm \sqrt{156}}{2}$$

$$= \frac{-12.49}{2} = -6.245$$

So larger one mod we get

$$\cancel{D} = -21 \quad x_3 = x_2 = \frac{-2c}{b - \sqrt{b^2 - 4ac}}$$

$$= 2 = \frac{2x - 3}{-21.7980}$$

$$= 2 \neq \frac{6}{21.7980}$$

$$= \underline{\underline{1.7247}}$$

Iteration 2.

Here we consider the parabola passing through $(x_1, f(x_1))$ $(x_2, f(x_2))$ $(x_3, f(x_3))$

$$x_0 \quad x_1 \quad x_2 \Rightarrow x_3$$

$$\boxed{x_1} \quad \boxed{x_2} \quad \boxed{x_3} \Rightarrow x_4$$

$$\begin{matrix} 1 & 2 & 1.7427 \\ c_2 & f(x_3) & = (1.7427)^3 - 7x(1.7427)^2 + 6x(1.7427) + 5 \\ & & = \cancel{-0.3441} - 0.3437 \end{matrix}$$

$$f(x_0) = f(x_1) = 5$$

$$f(x_1) = f(x_2) = -3$$

function (output args), untitled (input args).

$$\begin{bmatrix} x_0 (x_1 - x_3)^2 & x_1 - x_3 \\ (x_2 - x_3)^2 & (x_2 - x_3) \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} f(x_1) - c \\ f(x_2) - c \end{bmatrix}$$

$$\begin{bmatrix} (1 - 1.7427)^2 & (1 - 1.7427) \\ (2 - 1.7427)^2 & (2 - 1.7427) \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 5 + 0.3437 \\ -3 + 0.3437 \end{bmatrix}$$

$$\begin{bmatrix} 0.5252 & -0.7247 \\ 0.0758 & 0.2753 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 3.3437 \\ \cancel{3.3437} \\ \cancel{2.6563} \\ 2.6563 \end{bmatrix}$$

$$a = -2.2753$$

$$b = 9.0225 //$$

$$x_4 = x_3 = \frac{-b}{2 \pm \sqrt{b^2 - 4ac}} = \cancel{x_4 = x_3}$$

$$= 1.6862$$

$$1) b + \sqrt{b^2 - 4ac}$$

$$= \frac{-9.0225 + \sqrt{(-9.0225)^2 - 4(-2.2753)(-0.3437)}}{2}$$

$$= \cancel{-17} \quad 0.1750$$

$$2) b - \sqrt{b^2 - 4ac}$$

$$= \frac{-9.0225 - \sqrt{(-9.0225)^2 - 4(-2.2753)(-0.3437)}}{2}$$

$$= -17.8700$$

$$17.8700 > 0.1750$$

Wise